WATERMARK FORGERY USING AVERAGE-BASED COPY ATTACK

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GitHub Repository:- `https://github.com/AhireRohit/TML_Assignment_4`

1 INTRODUCTION

In this assignment, the goal was to forge invisible watermarks on clean target images. We were given 200 clean target images and 8 groups of watermarked source images. Each group, from `WM_1` to `WM_8`, contained 25 images watermarked with the same hidden message, but the actual watermarking method was unknown. The task was to slightly change the clean target images so that the watermark detector would detect the correct watermark, while the images should still look almost same to the original clean images. Since the final score depends on both watermark detection and LPIPS image quality, the attack needs to be strong enough but not too visible.

2 METHOD

Our final method is based on a simple average-copy idea. The main assumption is that all images inside one watermark folder share the same hidden watermark signal, while the visible image content is different. Because of this, averaging the watermarked source images can reduce image-specific content and keep some of the common watermark signal.

For each watermark group `WM_i`, we computed the average watermarked source image:

$$A_i = \frac{1}{25} \sum_{j=1}^{25} W_{i,j},$$

where $W_{i,j}$ is the j-th watermarked source image from group i. For a clean target image x, we used the average source image to create the main perturbation:

$$\Delta_{\text{avg}} = \lambda_i (A_i - x).$$

This means that the target image is moved slightly towards the average watermarked image of the correct group. We kept λ_i small, because using a large value makes the image visually worse and increases the LPIPS penalty.

We also used a small high-frequency residual component. For this, each watermarked source image was denoised using non-local means denoising. Then we subtracted the denoised image from the original watermarked image. This gives a weak residual pattern:

$$R_i = \text{trimmed\_mean}_j (W_{i,j} - D(W_{i,j})),$$

where $D(\cdot)$ is the denoising operation. We used a trimmed mean instead of a normal mean to reduce the effect of outlier image content.

The final forged image was generated as:

$$\hat{x} = \text{clip}(x + \Delta_{\text{avg}} + \beta_i R_i).$$

The best global setting used $\lambda = 0.16$, residual strength $\beta = 2.0$, low clipping threshold 30, total clipping threshold 40, and texture mask strength 0.12. The texture mask was used so that changes are slightly stronger in textured regions and weaker in smooth regions, because smooth regions show artifacts more easily.

After getting the best global result, we tried per-watermark ablation. We found that `WM_1` worked better with a stronger average-mix setting. For `WM_1`, we used $\lambda = 0.20$, residual strength $\beta = 1.0$, low clipping threshold 32, and total clipping threshold 42. The final submission therefore uses the global setting for most watermark groups and uses the stronger setting only for the `WM_1` batch.

3 RESULTS

We first tested a residual-only method using non-local means denoising. This reached a public leaderboard score of 0.387863. Increasing the residual strength did not improve the score, probably because the visual quality became worse and cancelled the gain in detection strength.

The average-mix method gave the main improvement. With $\lambda = 0.08$ and residual strength 4.0, the score increased to 0.412935. With $\lambda = 0.12$ and residual strength 3.0, the score improved to 0.436932. The best global setting, using $\lambda = 0.16$ and residual strength 2.0, reached 0.442007.

Then we replaced only the `WM_1` target batch with the stronger $\lambda = 0.20$, $\beta = 1.0$ variant. This gave the best public leaderboard score:

$$\mathbf{0.444834}.$$

Replacing `WM_2` with the same stronger variant did not improve the score further. So the final result keeps only the `WM_1` replacement.

We also tried other approaches such as stronger residual extraction, paired source residual mixing, t-statistic aggregation, nearest-source mixing, DCT and FFT based copying, a learned denoiser, and a surrogate CNN detector. These methods did not improve the public leaderboard score. From our experiments, the simple average-mix attack was the most useful direction.

4 CONCLUSION

This assignment shows that invisible watermarking can be weak against copy attacks when an attacker has several examples with the same hidden message. Even without knowing the watermarking algorithm, averaging watermarked examples can reveal a transferable signal. By adding a small part of this signal to clean target images, it was possible to cause false watermark detection while keeping the images visually close to the originals.

This matters in real systems because watermark detectors may be used for ownership claims, image provenance, or checking whether an image came from a specific model. If an attacker can forge such a watermark, then an unrelated image could be falsely linked to a model or owner. This can make watermark-based evidence less reliable. So watermarking systems should be tested against copy attacks, avoid reusing the same message too much, and should be combined with other ways of proving image origin.